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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Applied Cryptography, Post-Quantum Cryptography, Computer Networks, Operating Systems, Distributed Systems, Design and Analysis of Algorithms II

PROJECTS

mldsa-auth | C11, liboqs, libsodium, SQLite, CMake

github.com/happyc0der/mldsa-auth

- Post-quantum mutual auth and session protocol in C11: ML-DSA-65 signatures over hybrid X25519 + ML-KEM-768, ChaCha20-Poly1305 records.
- 0.45 ms median handshake, 736 MiB/s record sealing; ASan/UBSan CI on Linux and macOS, 12 fuzz targets, 192 mutations nightly.
- Built mldsa-authd on it: login codes, device enrolment, revocation, key rotation, recovery codes; ProVerif model of the handshake.

TideFit | TypeScript, Next.js 16, React 19, Supabase, Vercel

github.com/UtkarshMitta/tide-fit

App by Utkarsh Mittal; the audit, upgrade, tests, CI and launch are mine.

- Security audit of a Next.js trip planner: 15 findings confirmed on the running app (OAuth CSRF, open Supabase policy, secret in a plaintext cookie), 13 fixed.
- Upgraded Next 14 to 16 and React 18 to 19 by hand; npm audit 6 vulnerabilities to 0, tests 0 to 45, CI deploying main to Vercel.
- Launched at tide-fit.vercel.app: private Blob storage, edge rate limit verified at 20 requests then 429, Strava and Google sign-in verified live.

One-time-pad allocation (team of 3) | Python, pytest, ruff, GitHub Actions

github.com/happyc0der/cryptography-project-1

Team of three; the chunk-reserve protocol, proofs, simulator, tests, CI and report are my 2026 rewrite.

- Wait-free one-time-pad allocation for m broadcasting parties with d messages in flight; proof of no reuse, exact worst-case waste $d(2m-1) - m + (n \bmod d)$.
- Worst-case waste at $m=5, d=5, n=5,000$ cut from 4,000 pads (static partition) to 36, with zero blocked sends across the adversarial grid.
- Found two defects in the course handout's own two-party protocol, each pinned by a test; 310 tests, CI reruns the evaluation and diffs the results.

Hawk signature study | Python, NumPy, SymPy, Flask, pytest

github.com/happyc0der/hawk-signature-study

- Study of Hawk, the lattice signature scheme withdrawn from NIST standardisation in July 2026: 13-page report, talk, Flask keygen/sign/verify API and demo page.
- Six correctness fixes to the vendored hawk-py reference, with regression tests; one made Hawk-256 verification about 300x faster (6.04 s to 0.02 s).
- Documented the break beside the unedited report: key recovery reduced to SVP in dimension $n/2 + 1$, Hawk-512 from 2^{150} to 2^{108} .

COMPETITIONS

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Security: libsodium, liboqs, ML-DSA, ML-KEM, X25519, ChaCha20-Poly1305, OAuth 2.1, libFuzzer, ASan/UBSan, threat modeling

Systems: CMake, gRPC, SQLite, WebAssembly, Android (Jetpack Compose, Room), Next.js, React, FastAPI, Node.js, REST APIs, MCP

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale

ACTIVITIES

- Treasurer, OWASP Student Chapter, IIIT Delhi
- Attended the first International Quantum Communication Conclave, New Delhi, 2023